

MENG WANG

Department of Civil, Architectural and Environmental Engineering
The University of Texas at Austin, USA
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[Google Scholar Profile](#) | [Personal Website](#)

RESEARCH INTERESTS

My research centers on molecular interactions and electrolyte thermodynamics in aqueous systems, with a focus on advancing water and energy sustainability. I investigate how interactions among ions, water, polymer coatings, and mineral surfaces govern speciation, sorption, and transport, ultimately controlling chemical reactions in aqueous and interfacial environments. By integrating spectroscopy and scattering, thermodynamic and computational modeling, and multiscale experiments, I connect molecular-scale interactions to macroscopic processes in membrane-based separations, critical mineral recovery, and scaling and fouling control at the water–energy nexus.

Research keywords: ion association and solvation; interfacial chemistry; ion transport; polymer-functionalized interfaces; predictive thermodynamic modeling; sustainable separations and resource recovery.

EDUCATION

- | | |
|---|------------------------------------|
| University of Houston
Ph.D., Environmental Engineering | <i>Houston, TX</i>
2023 |
| <ul style="list-style-type: none">Dissertation: "Mechanistic Study on Polymer Coatings to Control Mineral Scaling in Desalination Treatment" | |
| Georgia Institute of Technology
M.S., Computer Science | <i>Atlanta, GA</i>
2026 |
| China University of Petroleum, Beijing
M.S., Petroleum Engineering | <i>Beijing, China</i>
2018 |
| <ul style="list-style-type: none">Thesis: "Numerical Study of Proppant Transport with Supercritical CO₂ in Fracture" | |
| Northeast Petroleum University
B.E., Petroleum Engineering | <i>Heilongjiang, China</i>
2015 |
| <ul style="list-style-type: none">Thesis: "Hydraulic Fracturing Model Featuring Initiation Beyond the Wellbore Wall for Directional Well in Coal Bed" | |

RESEARCH EXPERIENCE

- | | |
|--|--------------------------|
| The University of Texas at Austin
Postdoctoral Researcher Advisor: Dr. Lynn Katz | <i>08/2023 – Present</i> |
| <ul style="list-style-type: none">Developed a molecular transport framework linking ion association, hydration, and thermodynamic nonideality to aqueous speciation, sorption, and transport.Established complementary Raman and dielectric spectroscopy methods to quantify ion pairing and ion–solvent dynamics in sulfate electrolytes containing mono-, di-, and trivalent cations (e.g., Li⁺, Mg²⁺, and La³⁺).Developed an open-source computational pipeline (dielectric analyzer) for dielectric-spectrum fitting and relaxation-process deconvolution;Resolved discrepancies in sulfate ion-pair stability constants across measurement methods and applied the validated constants to real-world water matrices, quantifying the effect of ion pairing on sulfate speciation.Developed a spectroscopically constrained extended Pitzer model that incorporates ion-pairing equilibria, predicts activity coefficients of charged ion pairs, and reveals systematic bias in conventional models that neglect ion pairing.Contributed to a pilot-scale produced-water separation project by evaluating ceramic and hollow-fiber membrane performance through solution-chemistry analysis and surface characterization.Authored 4 first-author manuscripts (1 under review, 2 ready for submission, and 1 in preparation) and delivered three conference presentations. | |

University of Houston

08/2018 – 05/2023

Graduate Research Assistant | Advisor: Dr. Debora F. Rodrigues

- Developed a mechanistic framework linking polymer chemistry and interfacial hydration to resistance to organic fouling, biofouling, and scaling (e.g., gypsum) in reverse-osmosis (RO) systems.
- Performed cross-flow RO experiments on polymer-coated membranes, characterizing fouling behavior and mineral scaling at organic–water interfaces to evaluate membrane antifouling and anti-scaling performance.
- Integrated membrane-performance measurements with surface and scattering characterization to establish structure–scaling relationships for antifouling and anti-scaling interface design.
- Published 4 first-author and 3 co-authored papers and presented findings at two professional meetings.

Argonne National Laboratory—Advanced Photon Source (APS)

01/2019 – 11/2022

Visiting Student (concurrent with Ph.D. research at University of Houston)

- Served as the primary writer for two General User Proposals that secured 36 total shifts of beamtime at APS.
- Executed *in situ* SAXS/GISAXS and *ex situ* GIWAXS experiments at 12-ID-B to quantify early-stage mineral nucleation, growth, and crystallinity on polymer-coated interfaces.
- Processed and modeled X-ray scattering datasets using MATLAB, Igor Pro, and Irena SAS; the results contributed to a publication in *Water Research*.

China University of Petroleum, Beijing

09/2015 – 07/2018

Graduate Research Assistant | Advisor: Dr. Haizhu Wang

- Developed computational fluid dynamics (CFD) models of supercritical CO₂–proppant transport in fractures and used them to identify design parameters governing particle placement in waterless fracturing.
- Co-developed experimental and drilling devices, resulting in two granted Chinese patents.
- Published 1 first-author and 6 co-authored papers on supercritical CO₂ transport, fracturing, and subsurface applications.

Northeast Petroleum University

02/2014 – 07/2015

Undergraduate Research Student | Advisor: Dr. Yuwei Li

- Developed a coal-bed fracture-initiation model and wrote a C/C++ simulator to predict fracture-initiation pressure along coal cleats near wellbores during hydraulic fracturing.
- Published the model in the *Journal of Geophysics and Engineering*.

PUBLICATIONS

Manuscripts Under Review and in Preparation:

4. **M. Wang**, Z. Huang, J. Cui, N. Vinh, B. Freeman, and L. Katz. Spectroscopic Insights into Ion Pairing in Sulfate Electrolytes: Implications for Sorption in Aqueous Systems. *Environmental Science & Technology* (2026). [Under review]
3. **M. Wang** and L. Katz. Incorporating Ion Pairing into an Extended Pitzer Model for Activity Coefficients. *Chemical Engineering Journal* (2026). [In preparation]
2. **M. Wang**, J. Cui, N. Vinh, and L. Katz. Unraveling Ion Pairing and Hydration Structure of Sulfate Salts in Aqueous Solution: From Monovalent to Trivalent Ions. *Environmental Science & Technology* (2026). [In preparation]
1. **M. Wang**, R. Jacovone, L. Faria, X. Zuo, R. O'Hara, A. Asatekin, and D. Rodrigues. Zwitterion Loading Tunes Heterogeneous Gypsum Nucleation on Amphiphilic Copolymer Coatings. *Water Research* (2026). [In preparation]

Peer-Reviewed Journal Articles:

21. R. Jacovone, **M. Wang**, J. Soares, S. Sakata, and D. Rodrigues. Investigating the Influence of Polymaleic Acid on Reverse Osmosis Membrane Performance to Mitigate Gypsum Scaling. *Desalination* 619 (2026): 119544. <https://doi.org/10.1016/j.desal.2025.119544>.

20. **M. Wang**, M. Sarma, S. Lounder, A. Mondal, L. Muthusamy, G. Koley, A. Asatekin, and D. Rodrigues. Organic Fouling on Zwitterionic Amphiphilic Copolymers: Implications in Biofouling. *ACS Applied Materials & Interfaces* 17(20) (2025): 30149–30160. <https://doi.org/10.1021/acsami.5c07057>.
19. **M. Wang**, X. Zuo, R. Jacovone, R. O'Hara, A. Mondal, A. Asatekin, and D. Rodrigues. Influence of Zwitterionic Amphiphilic Copolymers on Heterogeneous Gypsum Formation: A Promising Approach for Scaling Resistance. *Water Research* 266 (2024): 122439. <https://doi.org/10.1016/j.watres.2024.122439>.
18. **M. Wang**, H. Nguyen, S. Lounder, A. Asatekin, and D. Rodrigues. Calcium Sulfate Formation on Different Zwitterionic Amphiphilic Copolymer Substrates for Salt Water Treatment. *ACS Applied Polymer Materials* 4(10) (2022): 7090–7101. <https://doi.org/10.1021/acsapm.2c01014>.
17. H. Arabaghian, **M. Wang**, J. Ordóñez, and D. Rodrigues. Zwitterionic Polymers in Biofouling and Inorganic Fouling Mechanisms. In *60 Years of the Loeb-Sourirajan Membrane* (2022): 33–70. <https://doi.org/10.1016/B978-0-323-89977-2.00013-0>. [Book chapter; co-first author]
16. **M. Wang**, B. Cao, Y. Hu, and D. Rodrigues. Mineral Scaling on Reverse Osmosis Membranes: Role of Mass, Orientation, and Crystallinity on Permeability. *Environmental Science & Technology* 55(23) (2021): 16110–16119. <https://doi.org/10.1021/acs.est.1c04143>.
15. A. Ansari, J. Peña-Bahamonde, **M. Wang**, D. Shaffer, Y. Hu, and D. Rodrigues. Polyacrylic Acid-Brushes Tethered to Graphene Oxide Membrane Coating for Scaling and Biofouling Mitigation on Reverse Osmosis Membranes. *Journal of Membrane Science* 630 (2021): 119308. <https://doi.org/10.1016/j.memsci.2021.119308>.
14. F. Chen, Y. Wang, R. Bi, Y. Pan, and **M. Wang**. Molecular Insights into CO₂-Enhanced Hydrocarbon Recovery and Its Sequestration in Multiscale Shale Reservoirs. *Chemical Engineering Journal* 498 (2024): 155913. <https://doi.org/10.1016/j.cej.2024.155913>.
13. Y. Wang, Q. Sun, F. Chen, and **M. Wang**. Multiscale Model for Hydrogen Transport and Storage in Shale Reservoirs. *SPE Journal* 29(6) (2024): 3238–3264. <https://doi.org/10.2118/219472-PA>.
12. T. Ge, **M. Wang**, F. Li, F. Min, and M. Zhang. Study and Spectroscopy Characterization of Carbon and Oxygen Structures of Gaoyang Coking Coal. *Journal of China Coal Society* 46(3) (2021): 1024–1031. <https://doi.org/10.13225/j.cnki.jccs.2019.1693>.
11. H. Wang, G. Li, Z. Shen, Z. He, Q. Liu, B. Zhu, Y. Wang, and **M. Wang**. Expulsive Force in the Development of CO₂ Sequestration: Application of SC-CO₂ Jet in Oil and Gas Extraction. *Frontiers in Energy* 13(1) (2019): 1–8. <https://doi.org/10.1007/s11708-017-0458-6>.
10. **M. Wang**, H. Wang, G. Li, B. Yang, Y. Zheng, and Q. Lu. Numerical Study of Proppant Transport with Supercritical CO₂ in Fracture. *China Petroleum Machinery* 46(11) (2018): 72–78, 84. <https://doi.org/10.16082/j.cnki.issn.1001-4578.2018.11.012>.
9. H. Wang, **M. Wang**, B. Yang, Q. Lu, Y. Zheng, and H. Zhao. Numerical Study of SC-CO₂ and Proppant Transport in Different Geometrical Fractures. *Greenhouse Gases: Science and Technology* 8(5) (2018): 898–910. <https://doi.org/10.1002/ghg.1803>.
8. H. Wang, G. Li, Z. He, S. Tian, **M. Wang**, B. Yang, Q. Lu, and L. Weng. Experimental Investigation on Abrasive Supercritical CO₂ Jet Perforation. *Journal of CO₂ Utilization* 28 (2018): 59–65. <https://doi.org/10.1016/j.jcou.2018.09.018>.
7. H. Wang, G. Li, Z. He, Z. Shen, X. Li, Z. Zhang, **M. Wang**, B. Yang, Y. Zheng, and L. Shi. Analysis of Rock Fracture Mechanism Caused by Supercritical CO₂. *Rock and Soil Mechanics* 39(10) (2018): 3589–3596. <https://doi.org/10.16285/j.rsm.2017.0071>.
6. H. Wang, G. Li, L. Zhao, Y. Wang, Y. Liu, Q. Liu, **M. Wang**, F. Gao, and M. Cai. Throttle Characteristics of Multi-Stage Circumfluence Nozzle During the Separate-Layer Injection of CO₂. *Petroleum* 4(2) (2018): 187–197. <https://doi.org/10.1016/j.petlm.2017.08.003>.
5. Y. Wang, J. Yuan, Z. Wang, Y. Yin, **M. Wang**, and W. Wang. Experimental Study on the Effect of Mechanical Vibration on Cement Stone Strength. *Natural Gas Exploration and Development* 41(1) (2018): 96–100.

4. H. Wang, G. Li, Z. He, Z. Shen, **M. Wang**, and Y. Wang. Mechanism Study on Rock Breaking with Supercritical Carbon Dioxide Jet. *Atomization and Sprays* 27(5) (2017): 383–394. <https://doi.org/10.1615/AtomizSpr.2017017174>.
3. X. Li, G. Li, H. Wang, S. Tian, X. Song, P. Lu, and **M. Wang**. A Unified Model for Wellbore Flow and Heat Transfer in Pure CO₂ Injection for Geological Sequestration, EOR, and Fracturing Operations. *International Journal of Greenhouse Gas Control* 57 (2017): 102–115. <https://doi.org/10.1016/j.ijggc.2016.11.030>.
2. Y. Li, D. Jia, **M. Wang**, J. Liu, C. Fu, X. Yang, and C. Ai. Hydraulic Fracturing Model Featuring Initiation Beyond the Wellbore Wall for Directional Well in Coal Bed. *Journal of Geophysics and Engineering* 13 (2016): 536–548. <https://doi.org/10.1088/1742-2132/13/4/536>.
1. X. Li, G. Li, H. Wang, P. Lu, S. Tian, R. Yang, and **M. Wang**. A Coupled Model for Predicting Flowing Temperature and Pressure Distribution in Drilling Ultra-Short Radius Radial Wells. *IADC/SPE Asia Pacific Drilling Technology Conference* (2016): SPE-180597-MS. <https://doi.org/10.2118/180597-MS>. [Conference proceeding]

GRANTED PATENTS

2. H. Wang, G. Li, J. Li, Z. Huang, **M. Wang**, B. Yang, Q. Lu. A wire-driven drilling device. Patent No. CN106703720B, China University of Petroleum, Beijing, 2018.
1. H. Wang, G. Li, **M. Wang**, S. Tian, Z. Huang, B. Yang, Q. Lu, L. Weng. Supercritical CO₂ sand-carrying simulation experimental device and system in fractures. Patent No. CN108590620B, China University of Petroleum, Beijing, 2020.

PRESENTATIONS

Oral Presentations:

5. **M. Wang**, B. Freeman, L. Katz, 2026. Incorporating ion pairing into an extended Pitzer model for activity coefficients: implications for ion exchange membrane modeling. *International Congress on Membranes and Membrane Processes (ICOM)*, San Antonio, TX.
4. **M. Wang**, J. Cui, N. Vinh, Z. Huang, B. Freeman, L. Katz, 2025. Comprehensive analysis of ion pairing and solvation in monovalent sulfate salts: insights from Raman and dielectric spectroscopy. *American Chemical Society Spring 2025 Meeting*, San Diego, CA.
3. **M. Wang**. Mineral Scaling on Reverse Osmosis Membranes: Role of Mass, Orientation, and Crystallinity on Permeability. Invited seminar, Environmental and Water Resources Engineering (EWRE), The University of Texas at Austin, Austin, TX, February 15, 2024.
2. **M. Wang**, H. Nguyen, S. Lounder, A. Asatekin, D. Rodrigues, 2021. Calcium sulfate formation on different zwitterionic amphiphilic copolymer substrates: implications for developing anti-scaling surfaces. *ACS Southwest Regional Meeting*, Austin, TX.
1. **M. Wang**, B. Cao, Y. Hu, 2019. Effects of organic macromolecules on gypsum scaling of reverse osmosis membranes. *Water Quality Technology Conference*, Dallas, TX.

Poster Presentations:

3. L. Katz, **M. Wang**, Z. Bajalan, Z. Huang, J. Cui, N. Vinh, B. Freeman, G. Henkelman, 2026. The role of ion pairing in membrane separations. *Gordon Research Conference on Environmental Sciences: Water*, Holderness, NH.
2. **M. Wang**, Z. Huang, J. Cui, N. Vinh, B. Freeman, L. Katz, 2025. Quantifying ion pairing in monovalent sulfate salts: implications for ion exchange membrane modeling. *AEESP 2025 Research and Education Conference*, Durham, NC.
1. Z. Huang, Z. Bajalan, **M. Wang**, R. Sujunani, G. Henkelman, L. Katz, B. Freeman, 2025. Effect of Ion Pairing on Salt Transport in Ion Exchange Membranes (IEMs). *The 2025 EFRC-Hub-CMS-CCS Principal Investigators' Meeting*, Rockville, MD, August 2025.

HONORS & AWARDS

8. **Early Career Postdoctoral Fellows Program**, The University of Texas at Austin, 2026
7. **American Chemical Society GEOC Early Career Award**, American Chemical Society, 2025
6. **Processes Outstanding Reviewer Award**, MDPI, 2025
5. **Graduate Student Travel Award**, ACS Southwest Regional Meeting, Austin, TX, 2021
4. **Graduate Tuition Fellowship**, University of Houston, Houston, TX, 2018–2023
3. **Robert Rocca Scholarship**, Tenaris China, 2017
2. **First-Class Scholarship**, China University of Petroleum, Beijing, 2017
1. **Outstanding Graduate Student**, China University of Petroleum, Beijing, 2016

FUNDING AND PROPOSAL EXPERIENCE

U.S. Department of Energy, Basic Energy Sciences (BES) — Energy Frontier Research Center 2026
Center for Materials for Extraction and Energy Transformations (MXET)

- **Role:** Proposal Contributor. Contributed to the development of the first research thrust by conducting preliminary data analysis, preparing figures, developing experimental methodologies, and synthesizing supporting literature.

Status: Under review.

PI: Lynn Katz.

Advanced Photon Source, Argonne National Laboratory — General User Proposal 2021
Effect of Zwitterionic Copolymer Surfaces with Structural Variability on Heterogeneous Nucleation and Growth of CaSO_4 (GUP No. 77339)

- **Role:** Primary proposal writer. Independently developed and wrote the proposal and experimental plan.

Award: 27 shifts of beamtime at APS 12-ID-B over three visits.

PI: Debora F. Rodrigues.

Advanced Photon Source, Argonne National Laboratory — General User Proposal 2020
Calcium Sulfate Scaling on the Zwitterionic Amphiphilic Copolymer Substrates: Implication for Developing Anti-Scaling Membrane Surfaces (GUP No. 72353)

- **Role:** Primary proposal writer. Independently developed and wrote the proposal and experimental plan.

Award: 9 shifts of beamtime at APS 12-ID-B for remote-access experiments.

PI: Debora F. Rodrigues.

National Natural Science Foundation of China (NSFC) — General Program 2018
Mechanism of Proppant Transport in Fractures by Supercritical CO_2 (Grant No. 51874318)

- **Role:** Proposal contributor. Contributed substantially to proposal writing, including literature review, identification of research gaps, formulation of research objectives, and experimental design.

Status: Funded.

PI: Haizhu Wang.

TEACHING EXPERIENCE

Co-instructor, Foundations of Environmental Engineering (**EVE 302**)

The University of Texas at Austin

01/2024–05/2024

- Co-designed and delivered laboratory activities connecting course concepts with experimental practice.
- Coordinated course examinations and supported consistent assessment of student learning.
- Provided direct laboratory instruction and guidance on experimental procedures and course concepts.

Teaching Assistant, Environmental Engineering (**CIVE 3331**)

University of Houston

08/2022–12/2022

- Delivered problem-solving lectures explaining homework methods and course concepts.
- Provided individualized support for students' conceptual and quantitative questions.
- Evaluated homework and examinations with the instructor to maintain consistent grading.

MENTORING EXPERIENCE

Undergraduate Students

- Denali Ibbotson, Colorado School of Mines, Summer 2019 (REU Program)
Current position: Ph.D. student, Texas A&M University
- Satvik Raghunathan, The University of Texas at Austin, Summer 2024
Current position: Undergraduate student, The University of Texas at Austin

Graduate Students

- Nikhil Khatri, University of Houston, Spring 2019
Current position: Environmental Engineer, Distinct Engineering Solutions, Inc.
- Vicky Agrawal, University of Houston, Fall 2019
Current position: Associate Consultant, WSP
- Raynara Maria Silva Jacovone, University of Houston, Fall 2021
Current position: Postdoctoral Researcher, Clemson University
- Murchana Sarma, University of Houston, Fall 2022
Current position: Ph.D. student, Clemson University
- Julia L. McGuire, The University of Texas at Austin, Fall 2023
Current organization: Chicago Consulting Company

ACADEMIC SERVICE

- **Journal Reviewer (105 peer reviews completed for 10+ academic journals)**
 - *npj Clean Water*
 - *Water Resources Research*
 - *Physics of Fluids*
 - *Energy Science & Engineering*
 - *Sustainability*
 - *Processes*
 - *Materials*
 - *Water*
 - *Energies*
 - *Fluids*
 - *Frontiers in Earth Science*
- **Editorial Service**
 - Guest Editor, *Water*, Special Issue on Membrane Fouling Control Strategies in Water Treatment 2024

PROFESSIONAL EXPERIENCE

Consultant Intern, Baker Tilly Beijing

Beijing, China | 05/2018–07/2018

- Supported internal audits, client proposal development, project analysis, and technical report preparation.

Research Intern, ConocoPhillips China

Beijing, China | 07/2017–09/2017

- Processed and analyzed oil-well production, drilling, and core-sample data, including data management, visualization, and sand-size distribution analysis.
- Evaluated wellbore designs and proppant size selection using drilling and formation sand-size data for the Penglai Oilfield.

TECHNICAL SKILLS

- **Membrane & Surface Characterization:**

Atomic force microscopy (AFM), scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS), X-ray diffraction (XRD), attenuated total reflectance Fourier-transform infrared spectroscopy (ATR-FTIR), quartz crystal microbalance with dissipation (QCM-D), zeta potential analysis

- **Synchrotron & Scattering Techniques:**

Grazing-incidence small-angle X-ray scattering (GISAXS/SAXS), grazing-incidence wide-angle X-ray scattering (GIWAXS), X-ray adsorption spectroscopy

- **Spectroscopy & Chemical Analysis:**

Raman spectroscopy, dielectric spectroscopy, inductively coupled plasma mass spectrometry (ICP–MS and ICP–OES), atomic absorption spectroscopy (AAS), ion chromatography, total organic carbon (TOC) analysis, UV–Vis spectroscopy

- **Computational & Thermodynamic Modeling:**

Geochemical modeling (Geochemist's Workbench), computational fluid dynamics (ANSYS Fluent, ICEM CFD, CFD-Post), thermodynamic modeling (MINTeq)

- **Programming & Data Science:**

Python, MATLAB, C/C++, Linux Shell, Git

- **Engineering & Design Software:**

AutoCAD, SolidWorks

REFERENCES

Debora F. Rodrigues, Professor

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Lynn E. Katz, Professor

Department of Civil, Architectural and Environmental Engineering, The University of Texas at Austin

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Benny D. Freeman, Professor

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Chemical and Biological Engineering, Tufts University

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Vinh Q. Nguyen, Associate Professor

Department of Physics, Virginia Tech

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